

Approximating Natural Language (Shannon's Method)

Basics of AI – Fall 2025

1. Introduction

Claude Shannon's 1948 paper "*A Mathematical Theory of Communication*" demonstrated that language is not random but follows statistical patterns. He introduced the idea that English can be modeled as a probabilistic process — where each character or word depends on the previous ones. This assignment recreates Shannon's classic text generation experiment using modern computing tools. By building n-gram models (zero-order to third-order), I generated artificial text for three authors — Jane Austen, Mark Twain, and Arthur Conan Doyle — and analyzed how linguistic structure improves with increased context.

2. Methodology

2.1 Data Collection and Preprocessing

- Texts used:
 - ✓ *Pride and Prejudice* – Jane Austen
 - ✓ *Adventures of Tom Sawyer* – Mark Twain
 - ✓ *Sherlock Holmes* – Arthur Conan Doyle
- All texts were cleaned using `starter_preprocess.py` (lowercasing, punctuation normalization, tokenization).
- Frequency tables created:
 - Characters: unigram, bigram, trigram
 - Words: unigram, bigram, trigram
 - Sentence length distribution (mean, standard deviation, histogram)

2.2 Model Implementation

- For each author, I created character-level and word-level n-gram models.
- **Character-level models implemented:**
 - **char-0:** random characters, equal probability
 - **char-1:** characters chosen by English frequency but independent
 - **char-2:** bigram model – $P(c_i | c_{i-1})$
 - **char-3:** trigram model – $P(c_i | c_{i-2}, c_{i-1})$
- **Word-level models:**
 - **word-1:** random words based on unigram frequency
 - **word-2:** bigram words ($P(w_i | w_{i-1})$)

- **word-3:** trigram words ($P(w_i | w_{i-2}, w_{i-1})$)
- A function added to generate sentences matching the author's sentence-length distribution.

3. Results and Observations

3.1 Quality Improvement from 0th to 3rd Order

Model	Example Output	Notes
char-0	XFQJ AZZPL DMKS	Pure noise, no English structure
char-1	TENE RIASL OAEH	More vowels, but meaningless
char-2	Ther an to be withe her sion	Realistic fragments like “th”, “her”
char-3	She could not believe that he was	Nearly readable English structure
word-1	the she and was very the	Real words but incoherent
word-2	she was not able to speak	Phrases begin to make grammatical sense
word-3	Elizabeth could not help smiling as she replied	Full sentence, strongly author-like

Conclusion: As the value of n increases, the output becomes more grammatical, coherent, and stylistically closer to the original author.

3.2 Author Style Recognition

- At **char-2** and **char-3**, we begin to see author-specific patterns:
 - Austen uses words like “Elizabeth”, “marriage”, “feelings”.
 - Twain generates “river”, “Aunt Polly”, “Tom ran” patterns.
 - Doyle often produces “Holmes”, “Watson”, “crime”, “deduction”.
 - True stylistic identity becomes clearly recognizable only from **word-3** onwards.
-

3.3 Anchor Word Integration

- The requirement was to generate text that *must include specific words* like "elementary, Watson, deduce".
- Strategy used:
 - Generate multiple candidate sentences.
 - Check if all anchor words appear.

- If not found after ~50 attempts → add a natural fallback sentence at end.
 - Challenge: Maintaining flow while forcing words. Higher n-values tend to exclude rare words, making anchor insertion harder.
-

4. Connection to Modern AI

N-gram Models

Modern Language Models (GPT, BERT)

Only depend on previous 1–2 tokens Use full attention over entire sentence/paragraph

No memory of long-range context Transformers capture global context via attention

Cannot understand meaning Use word embeddings and semantics

Small probability tables Trained on billions of parameters & tokens

Limitations of N-grams solved by Transformers:

- ✓ Long-range dependencies
 - ✓ Vocabulary generalization
 - ✓ Sentence coherence and meaning
 - ✓ Creativity and reasoning
-

5. Shannon's Insight Reflected

"Language can be approximated statistically." – Claude Shannon

This assignment proves Shannon's idea: even without understanding grammar or meaning, **statistical frequency alone can recreate the illusion of language**. By increasing n, the model captures more context, improving structure and realism. This is exactly the foundation upon which modern AI (like GPT-5) is built — but at scale, with deeper networks and far larger context windows.

6. Conclusion

- Character and word n-gram models can generate surprisingly realistic English.
 - Author style becomes recognizable at **word-3** level.
 - Anchor word integration is achievable but complex due to statistical limitations.
 - Modern AI models like GPT are direct descendants of Shannon's Markov chains — scaled with deep learning and massive datasets.
-

7. References

- Shannon, C. E. (1948). *A Mathematical Theory of Communication*.

- Project Gutenberg eBooks (Austen, Twain, Doyle).
- Chen & Goodman (1999) – *N-gram Smoothing Techniques*.